

EECS 2311 Team 10

Planning Document

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Original Plan Details

Our team would like to create a location sharing social platform meant for photographers of various levels to share and promote their photos and where they took them.

Vision Statement: FotoFinder

Project Overview

FotoFinder is a community-driven photography location-sharing platform designed to help photographers and hobbyists discover scenic spots for photography. It eliminates the guesswork and time-consuming travel involved in finding the perfect location by allowing users to pin and explore photography-friendly locations contributed by the community.

Target Users

The primary users of FotoFinder are:

- Professional photographers looking for new and unique photography spots.
- Hobbyist photographers who want to explore new places and improve their craft.
- Travelers and outdoor enthusiasts interested in capturing stunning landscapes.

Project Value & Differentiation

- **Convenience:** Users can find pre-vetted locations with high-quality image previews, eliminating the need to scout locations manually.
- **Community Engagement:** Users can rate locations, leave comments, and follow fellow photographers, creating an interactive and supportive environment.
- **Location Accuracy:** Integrates Leaflet Map API and Nominatim Geocoder API to provide precise coordinates and address details for each pinned location.
- **Cloud-Based Storage:** All user data, images, and location pins are securely stored in Firebase, ensuring fast and seamless access across devices.
- **Smart Filtering:** Users can search by categories (urban, nature, night photography, etc.), weather conditions, or user ratings to find the perfect spot.

Technical Implementation

- **Frontend:** Next.js or Vite with TypeScript, using the Leaflet Map API for map visualization.
- **Backend:** Firebase for user authentication, database management (RTDB), and image storage.
- **Testing:** Jest for unit and integration testing to ensure a robust and scalable system.

Success Criteria

FotoFinder will be considered successful if it:

- ❖ Attracts active users who pin locations and contribute high-quality images.
- ❖ Provides a smooth and intuitive user experience, allowing users to explore and filter locations effortlessly.
- ❖ Ensures secure and reliable authentication and data storage using Firebase.
- ❖ Encourages community engagement through interactions like comments, ratings, and follows.
- ❖ Implements efficient search and filtering capabilities to enhance usability.

By Humayun

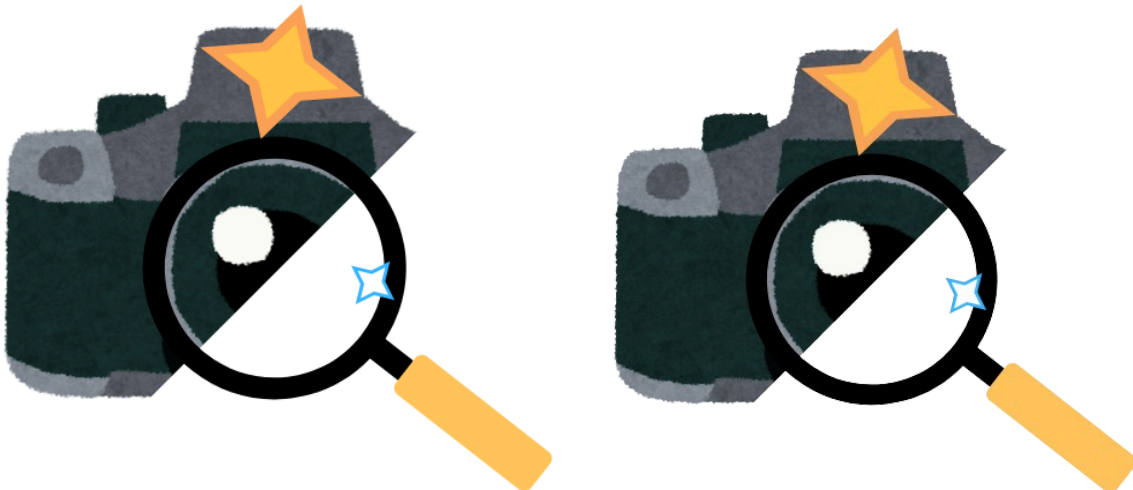
Initial Suggested Delegation of Tasks

- Humayun Ejaz:
 - Vision statement
- Alireza Hakim:
 - User stories
 - Big user stories
 - Creating Planning Document
- Johnson Han:
 - User stories
 - General code development
- Manoj Ravichandran:
 - General development
- Daniel Rusetski:
 - Video interview
 - Code environment setup
 - GitHub setup
- Umer Shaikh:
 - Frontend development

Application Logo

Initial Logo Design:

[Canva Link](#)



By Alireza

Technical Features to Add Based on Detailed User Stories

User Story Number	User Story	Technical Details
1	Map of Photos	Map with pins of the posted photos
2	Photo Uploads	Ability to upload posts that include the uploaded photo(s) and their location Also include camera specs of the camera that took the photo
3	Map Filter	Filter the photos on the left hand feed based on their theme
4	Security and IP Protection	Create a method so that uploaded photos cannot be stolen easily (possibly by reducing resolution, adding a watermark, etc.)
5	Custom Resolution Uploads	Allow users to choose the resolution that they upload their photos with.

5	Profile System	Allow users to sign up and create accounts that allow them to upload their photos
6	Review System	Allow users to rate posts and/or locations out of 5 stars.
7	Photo Feed	Have a left hand scrolling feed that shows the uploaded photos
8	Profile Following	User profiles should be able to follow and view each other, like other social media

Added User Stories

User Story Number	User Story	Technical Details
9	Detailed Filter and Search	Allow users to search for photos in a search bar, regarding tags and/or photo contents
10	Location Filter	Filter the photos and posts based on their location
11	Dark Mode	Allow users to use the website in dark mode

Coding Plans

1. Create directories for reusable components and pages. Don't need to add reactivity yet, but should be a single page application.
2. Create the canvas for the home page with HTML and Tailwind CSS.
Essentially just creating divs that match the layout of the concept screenshots sent. No styling necessary yet, but can be started.
3. Set up nominatim geocoder API and leaflet map after canvas is finished.
4. Set up backend (Firebase or AWS) and ensure a database can be queried and the frontend interfaces the backend.
5. Set up testing framework and run tests with non-persistent data.

By Johnson

Meeting Log Entries

Meeting 1 - 10:01 AM 2025-02-03

The TA suggested to incorporate features that will make our app unique compared to other social media-adjacent apps, so we should incorporate things that are unique to photographers (e.g. camera settings).

Potential unique features:

- Posts including the camera settings/specs that were used to take the picture
- Review system so people can rate recommended locations and users (kind of like Google Maps or Uber)
- User profile can include current camera and maybe a camera history (history of cameras the user has had)
- General recommendations/tips for camera settings
- Posts can include the raw file and the edited picture

General features:

- Dark mode
- Follower/following system

Meeting 2 - 9:49 AM 2025-02-24

ITR0 and ITR1 are done, just need to continue coding, especially our code foundation. All our planning documents are done and uploaded to the GitHub. Currently making plans to meet and catch up on any shortcomings in terms of the project's code.

Meeting 3 - 8:02 PM 2025-02-24

Coding todo list:

- Setup Map API/Basic UI (Leaflet)
- Setup FireBase
- Create and Connect Backend to Frontend
- Add posts to test pins throughout the map
- Add a separate Post feature
- Add user profiles and authentication through AWS
- Figma visuals for how we want the website to look like

Action Items:

- Create concepts of visuals on Figma
- Get familiar with relevant tools that we will use (codepen.io)

Meeting 4 - 9:45 AM 2025-03-03

Notes about UI elements:

- Map
 - Shows posts as map pins (group multiple nearby posts together?)
- Lefthand side-menu
 - Add post button
 - View post (happens when you click a map pin/post)
 - View feed button
 - Can be minimized to a search bar (like Google Maps)
 - Stays static
- User profiles
 - Display a scrolling feed of their posts
 - Lefthand menu shows user information (name, camera specs, etc)
- Darkroom Gallery (Social media feed)
 - Similar to Instagram explore page
 - Displays nearby photos
 - Clicking a photo expands it and shows its information (location, photographer, etc)

Meeting 5 - 10:04 AM 2025-03-24

- We mainly did a progress check-in with each other to prepare for the presentation
- We delegated the tasks to get ready for Delivery 2

Update on User Stories

User Story Number	User Story	Technical Details (Planned vs Implemented)
1	Map of Photos	Planned: Map with pins of the posted photos. Implemented: We have a functioning map that has each post as a pin that can be opened by the user to view the post.
2	Photo Uploads	Planned: Ability to upload posts that include the uploaded photo(s) and their location Also include camera specs of the camera that took the photo. Implemented: Users can upload one or more photos to post, along with a title and description along with the location in which it was taken. The specs of the used camera is scraped from the

		image's metadata.
3	Map Filter	Planned: Filter the photos on the left hand feed based on their theme. Implemented: Photos can be given a theme tag when uploaded and can then be filtered based on theme in the left hand feed.
4	Security and IP Protection	Planned: Create a method so that uploaded photos cannot be stolen easily (possibly by reducing resolution, adding a watermark, etc.). Implemented: Uploaded photos are automatically compressed, decreasing their resolution when compared to the main photo, discouraging IP theft.
5	Profile System	Planned: Allow users to sign up and create accounts that allow them to upload their photos. Implemented: Users can create accounts to upload photos, attaching their photos to their specific account.
6	Review System	Planned: Allow users to rate posts and/or locations out of 5 stars. Was not implemented
7	Photo Feed	Planned: Have a left hand scrolling feed that shows the uploaded photos. Implemented: All uploaded photos can be seen and scrolled through in the left hand feed.
8	Profile Following	Planned: User profiles should be able to follow and view each other, like other social media. Was not implemented
9	Detailed Filter and Search	Planned: Allow users to search for photos in a search bar, regarding tags and/or photo contents. Was not implemented
10	Location Filter	Planned: Filter the photos and posts based on their location. Implemented: Not explicitly implemented, deemed unnecessary since users can simply

		zoom in and out of the map to manually filter posts from desired locations.
11	Dark Mode	Planned: Allow users to use the website in dark mode. Implemented: The entire site is in dark mode by default.